

Windows Run Commands Reference Guide

Computer Science & Coding (Software) Reference Manual

Shortcut Verification: Press Win + R or right-click the Windows icon to open the Run utility.

This technical guide compiles core system diagnostics and engineering utilities to assist with hardware evaluation, full-stack software development workflows, and networking diagnostics.

1. System Performance & File Management Log

Run Command	Description & Practical Utility	Target Application
<code>resmon</code>	Tracks real-time utilization of CPU, Memory, Disk, and Network bandwidth. Crucial for identifying specific background processes exhausting system resources.	Resource Monitor
<code>explorer</code>	Serves as an alternate, command-based mechanism to open the native file system manager when taskbar shortcuts or graphical shell elements become unresponsive.	File Explorer
<code>cleanmgr</code>	Launches the automated system storage cleanup dialog to analyze and purge junk files, operational caches, and temporary system clutter from selected drives.	Disk Cleanup
<code>taskmgr</code>	Bypasses peripheral or shell inputs to immediately force-open the process manager, allowing administrators to terminate frozen applications and review hardware loads.	Task Manager

2. Advanced Administration & Diagnostics Log

Run Command	Description & Practical Utility	Target Application
<code>msconfig</code>	A comprehensive deployment manager to toggle diagnostic/selective startups, configure Windows environment parameters, or isolate drivers by booting into Safe Mode.	System Configuration
<code>rstrui.exe</code>	Directly opens the historical operating snapshot environment to quickly roll back critical OS states, registry baselines, or drivers following failed installations.	System Restore

Run Command	Description & Practical Utility	Target Application
mrt	Triggers the native Microsoft Malicious Software Removal Tool to execute on-demand background sweeps, targeting hidden malware threats independently of external antivirus suites.	Malware Removal Tool

3. Operating System & Hardware Verification Log

Run Command	Description & Practical Utility	Target Application
winver	Displays the essential micro-build specifications of the OS environment, revealing the exact edition (Home/Pro) and update version currently installed.	About Windows
msinfo32	Exposes a granular hardware and software inventory, mapping motherboard system models, exact processor architectures, current BIOS versions, and active Secure Boot profiles.	System Information
main.cpl	Direct control panel shortcut mapping straight to local peripheral hardware characteristics, including double-click sensitivity, pointer profiles, and tracking speeds.	Mouse Properties

4. Advanced Configuration & Networking Log

Run Command	Description & Practical Utility	Target Application
mstsc	Invokes the internal Microsoft Terminal Services Client to quickly establish direct remote graphical management connections to other host nodes on a shared local subnet.	Remote Desktop
gpedit.msc	Provides complete administrative access to the hierarchical rule registry to modify system permissions, restrict apps, and customize device-wide operating constraints.	Group Policy Editor
regedit	Provides direct programmatic access to the root database structure of the operating system to perform advanced system engineering modifications and low-level manual registry adjustments.	Registry Editor
powercfg.cpl	Opens legacy management profiles for device energy properties, controlling critical operating states like Hybrid Sleep, Fast Startup flags, and targeted performance profiles.	Power Options

Run Command	Description & Practical Utility	Target Application
<code>ncpa.cpl</code>	Directly loads the centralized physical and logical network adapter panel to easily view statuses, diagnose connection drops, or reconfigure IPv4/IPv6 address interfaces.	Network Connections

5. Accessibility & Software Management Log

Run Command	Description & Practical Utility	Target Application
<code>magnify</code>	Immediately launches the system accessibility screen-scaling tool to magnify selected focal display quadrants.	Magnifier
<code>appwiz.cpl</code>	Direct access route to system package configurations to clean up unwanted third-party desktop distributions or manually toggle key underlying Windows components (e.g., Hyper-V, IIS).	Programs and Features

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